

CHERNI VRAH, Bulgaria

WMO: 156130

Lat: 42.562N

Lon: 23.285E

Elev: 2292

StdP: 76.65

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.1	-17.2	-25.4	0.5	-11.9	-22.4	0.7	-10.9	34.1	-6.7	33.6	-6.3	12.1	320	1.029

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.5	17.3	11.0	15.9	10.5	14.7	10.1	12.4	15.1	11.6	14.2	10.8	13.5	5.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	11.1	13.5	10.5	10.4	12.7	9.6	9.9	11.9	42.5	15.1	40.0	14.3	37.8	13.6	18.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 28.3	24.1	19.8	DB	-21.7	20.5	2.4	1.3	-23.4	21.4	-24.8	22.2	-26.1	22.9	-27.9	23.8	(4)
(5)			WB	-21.9	15.4	2.4	1.5	-23.6	16.5	-25.0	17.4	-26.4	18.2	-28.2	19.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-7.2	-7.7	-5.4	-1.7	3.8	7.6	9.6	9.9	5.8	2.3	-1.7	-5.8	(6)	
	DBStd	7.57	4.74	4.89	4.65	3.92	3.78	3.85	3.44	3.14	3.75	4.38	5.23	4.70	(7)	
	HDD10.0	3448	532	496	478	352	193	92	50	40	133	240	351	491	(8)	
	HDD18.3	6386	790	729	736	602	450	323	272	261	376	497	601	749	(9)	
	CDD10.0	103	0	0	0	0	2	18	37	38	7	1	1	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	8.3	10.0	10.4	10.5	9.5	7.6	6.6	6.3	5.9	6.6	8.0	9.1	9.6	(14)	
(15) Precipitation	PrecAvg	720	47	46	54	61	89	85	68	60	62	57	41	49	(15)	
	PrecMax	1146	100	76	124	146	187	172	153	195	202	117	100	91	(16)	
	PrecMin	408	13	12	19	16	33	22	21	6	12	9	2	14	(17)	
	PrecStd	168	24	19	23	28	40	37	37	41	41	31	20	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	2.8	6.0	9.3	14.1	17.9	20.0	18.9	15.5	13.0	11.1	4.7	(19)	
		MCWB	-1.5	-1.9	2.1	5.6	8.6	11.6	10.7	11.4	9.9	7.8	3.9	-0.3	(20)	
	2%	DB	1.4	0.7	3.0	6.2	11.9	15.5	17.5	17.2	13.7	10.6	8.1	3.0	(21)	
		MCWB	-2.5	-2.1	0.0	2.6	7.0	10.4	10.9	11.0	9.2	6.3	1.8	-0.8	(22)	
	5%	DB	-0.3	-0.7	1.5	4.5	10.4	14.2	16.0	15.8	12.5	9.3	6.1	1.2	(23)	
		MCWB	-2.5	-2.4	-0.8	2.0	6.6	10.0	10.7	10.6	8.4	5.3	1.0	-1.0	(24)	
	10%	DB	-1.5	-1.9	0.4	2.9	8.9	12.8	14.6	14.5	10.9	7.8	4.3	-0.2	(25)	
		MCWB	-2.8	-2.9	-1.0	1.3	6.0	9.3	10.2	10.0	7.8	4.6	0.9	-1.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	-0.1	2.6	5.7	9.4	13.3	13.9	13.9	10.9	9.2	5.6	2.5	(27)	
		MCDB	1.3	0.4	4.9	8.8	12.4	16.1	16.8	16.7	14.0	11.6	8.7	2.9	(28)	
	2%	WB	-0.9	-1.3	0.9	3.5	8.1	11.7	12.3	12.3	9.9	7.7	4.0	0.9	(29)	
		MCDB	-0.2	-0.5	1.7	5.1	10.5	14.2	15.0	15.1	12.8	9.7	6.1	1.5	(30)	
	5%	WB	-1.9	-2.0	0.0	2.4	7.3	10.7	11.5	11.5	9.1	6.6	2.9	-0.3	(31)	
		MCDB	-1.1	-1.2	0.8	3.8	9.4	13.2	14.1	14.3	11.4	8.2	4.0	0.5	(32)	
	10%	WB	-2.8	-2.8	-0.8	1.5	6.5	9.7	10.8	10.6	8.2	5.7	1.7	-1.3	(33)	
		MCDB	-1.9	-2.1	0.2	2.7	8.3	12.2	13.4	13.4	10.2	6.8	3.3	-0.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	4.2	4.3	3.8	4.0	4.5	4.6	4.5	4.1	4.1	3.9	3.9	(35)	
		MCDBR	4.6	4.4	4.3	5.1	5.6	5.8	6.3	6.0	5.6	5.1	4.9	4.3	(36)	
		MCWBR	3.7	3.5	3.2	3.7	3.9	4.7	4.6	4.2	3.9	4.0	3.4	3.5	(37)	
	5% WB	MCDBR	3.9	4.3	3.9	4.5	5.0	5.0	5.4	5.4	5.2	4.3	4.5	3.5	(38)	
		MCWBR	3.6	4.0	3.4	3.6	3.9	4.8	4.6	4.4	4.0	3.7	3.6	3.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.241	0.260	0.290	0.329	0.325	0.322	0.323	0.329	0.312	0.285	0.262	0.243	(40)	
	taud		2.492	2.478	2.461	2.365	2.408	2.429	2.417	2.398	2.449	2.531	2.569	2.528	(41)	
	Ebn at Noon		931	961	962	939	947	947	942	927	923	916	898	898	(42)	
	Edh at Noon		70	84	98	117	115	113	113	111	98	79	65	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.62	2.45	3.46	4.44	5.29	6.01	6.24	5.64	4.16	2.90	1.86	1.37	(44)	
	RadStd		0.24	0.31	0.38	0.53	0.44	0.53	0.49	0.50	0.34	0.36	0.19	0.20	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-145	N/A	N/A	N/A

Nomenclature: See separate page